

Automated Moderation Using Paired Text Understanding

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Abstract

In this task, we leverage models that can identify whether a certain user comment breaches a particular community rule, according to the guidelines of Agile Community Rules Classification competition on Kaggle. The relationship between two textual inputs (comment and rule) must be jointly interpreted in this task, which is an example of a paired text categorization problem. In this research, we examine various modelling techniques, ranging from transformer-based architectures like RoBERTa and DistilBERT to LLM fine-tuning. Here, we experiment with different preprocessing, data augmentation, and feature generation on performance.

Introduction

Online community platforms with massive inflow of users have to rely on clearly defined moderating procedures. As these platforms expand, it becomes more and more difficult for moderators to manually assess user-generated content since they have to weigh enormous amounts of comments against intricate policy guidelines. Hence, creation of automated technologies that can support or partially replace manual moderation is of essence.

This real-world issue is the primary objective of the Agile Community Rules Classification competition on Kaggle, which offers a dataset of user comments together with detailed rule definitions. This problem is a paired text classification challenge where the model is given a rule and a remark, and it has to decide if the comment breaks the rule. In contrast to conventional single-input categorization, this configuration ensures that the model can comprehend the relationship between the semantics of the rule and the comment.

We investigate various natural language processing techniques in our solution, ranging from basic baselines to complex transformer models. We also look at performance-enhancing modelling, preprocessing, and data augmentation techniques. In the end, addressing this issue helps create automatic moderating solutions that are more reliable and scalable, promoting safer online communities.

Literature Survey

As online communities expand, the need for automated content moderation has grown, and several current methods approach moderation as a text categorization problem. However, more sophisticated methods consider the connection between a user's comment and the particular community guideline or rule rather than just identifying toxicity or spam in isolation. Such models are better suitable for real-world moderation tasks because of this paired configuration, which helps capture the nuanced nature of whether a comment genuinely breaches a specific guideline.

Siamese (dual-encoder) architectures have been developed to efficiently model sentence-pair or text-pair associations. Sentence-BERT (SBERT) (Reimers and Gurevych 2019) is a popular technique that creates rich, semantically relevant sentence embeddings by modifying the transformer BERT into a Siamese network. SBERT allows for quick and scalable similarity comparisons without the high computational expense of jointly encoding each pair. It achieves this by encoding each text independently and comparing them using cosine similarity.

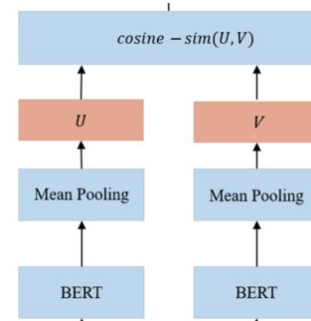


Figure 1: SBERT Architecture, adapted from (Zhou et al. 2023)

Model size and inference cost are important factors in real-world moderation systems. Lighter transformer models, such as DistilBERT (Sanh et al. 2019), provide a useful trade-off for various use cases by maintaining a significant portion of full-model performance while being quicker and more resource-efficient. When implementing moderating tools on a large scale, these methods are very attractive.

Preprocessing

Dataset Description

There are two primary text inputs in the dataset: rule, which represents the particular rule that is supposedly broken, and body, which represents the user's potentially rule-violating comment. There is a 2029-sentence corpus which differs greatly in length, style, and linguistic quality. The sentences consist of colloquialisms, acronyms, misspellings, irony, and jargon unique to a certain field. Significant noise is introduced into the classification task by these causes. In contrast, rule descriptions are more formal and structured, frequently outlining appropriate conduct, limitations, or content boundaries. The way these two text types interact is important because every rule-body pair may or may not be a violation.

```
row_id      2029
body        2029
rule        2029
subreddit    2029
positive_example_1  2029
positive_example_2  2029
negative_example_1  2029
negative_example_2  2029
rule_violation  2029
dtype: int64
```

Figure 2: Total Counts of Rows and Columns in the Raw Dataset

The dataset contains four auxiliary fields: `positive_example_1`, `positive_example_2`, `negative_example_1`, and `negative_example_2`. These fields include carefully chosen cases that demonstrate the practical applications of each rule. Positive examples demonstrate comments that violate that particular rule of the Subreddit, and the negative examples comply with the rule. These accurately chosen examples provide specific instances of acceptable and unacceptable behaviour. These examples serve as helpful interpretive guidelines, elucidating the intended meaning and reducing ambiguity around borderline circumstances because many rules can be abstract, subjective, or extremely contextual.

Dataset Metrics

The training set only contains examples from two classes of rules. However, the model must be able to predict other hidden classes of rules mentioned in the test dataset as well. The positive and negative samples per rule are distributed in the following manner as shown in figure 3.

There are 100 unique subreddits, with unequal distribution of rule violation examples. The top 10 subreddits with most rule violation are as described in figure 4.

Preprocessing Pipeline

Data Cleaning and Text Normalization Some textual fields in the dataset are systematically cleaned at the start of the preprocessing pipeline. After being converted to string format, all comment body and rule text fields are run via

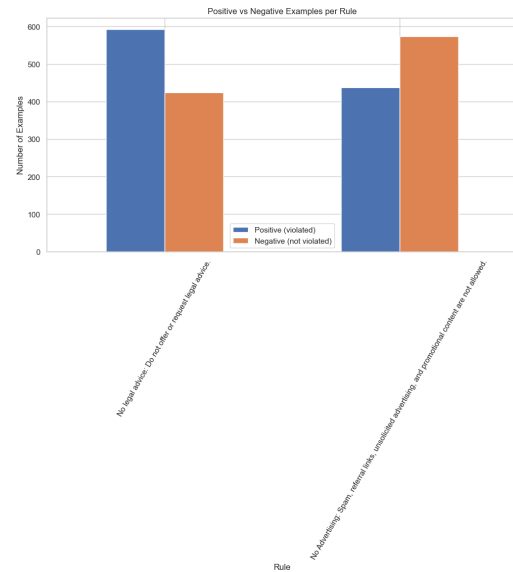


Figure 3: Positive vs Negative Samples per Rule

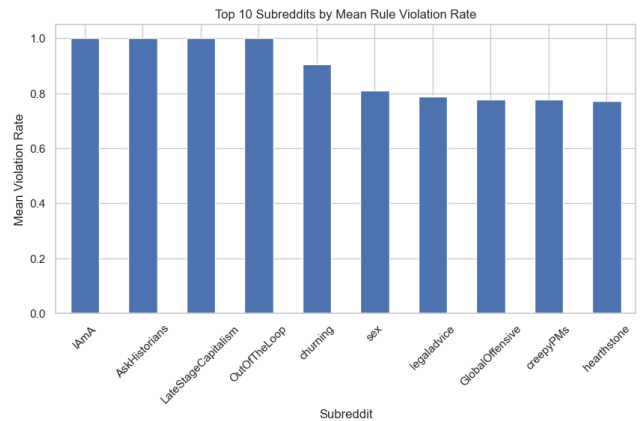


Figure 4: Top 10 Subreddits from Mean Rule Violation

a custom text-cleaning function, which performs tasks like general text sanitization, punctuation removal, lowercase, and white-space normalization. By standardizing the language before modelling, this stage minimizes vocabulary fragmentation and reduces noise. To preserve label consistency, rows with missing labels in the `rule_violation` field are removed.

Example: FindSexToday . com - free girls for sex world-wide 44XtPDKtcnDY30! [SEP] No Advertising: Spam, referral links, unsolicited advertising, and promotional content are not allowed.

Combining Comment and Rule Text into Unified Input

The cleaned text fields are combined into a single combined field named `combined_text` in order to explicitly condition the model on both the user comment and the associated community rule. This is accomplished using a custom function that concatenates the rule and comment us-

ing a structured separator token ([SEP]). Models, especially Transformers, can jointly reason about the subject of the comment versus the rule under evaluation thanks to this design.

Data Augmentation for Better Class Balance To improve the dataset and minimize class imbalance, an augmentation procedure is used before cleaning. To ensure consistent formatting, these augmented rows are subsequently subjected to the same cleaning and combination procedures. By increasing linguistic diversity, patterns of sentences and decreasing overfitting on a small number of rule-violating samples, this step improves the robustness of downstream classifiers.

```
Data augmentation complete!
Original rows (body only): 2629
Augmented rows (body + examples): 8318
New rows added: 6281

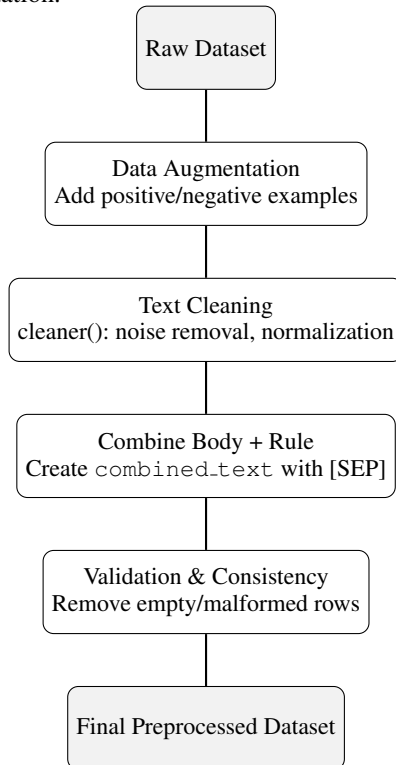
Label distribution:
rule_violation
1 4229
0 4089
Name: count, dtype: int64

Columns in new df (examples removed): ['body', 'rule', 'subreddit', 'rule_violation', 'source']

Applied combine_comment_rule to augmented dataset
Final columns after augmentation and unique comment-rule pair combination: ['body', 'rule', 'subreddit', 'rule_violation', 'source', 'combined_text']
Total rows after cleaning: 8318
```

Figure 5: Increase in Dataset after Augmentation

Validation of Newly Generated Data The pipeline checks for file encoding problems, proper labelling formats, empty or missing combined_text rows, and newline normalization.



Models

In the first approach, we use three variations of the BERT architecture; DeBERTa, DistilRoBERTa, and DistilBERT,

to fine-tune Transformer-based encoders in a supervised manner. These models are cutting-edge pre-trained language models that use deep contextual representations to encode text. Although DistilRoBERTa and DistilBERT offer lightweight distillation-based alternatives that maintain language understanding performance at reduced computational cost, DeBERTa (Decoding-Enhanced BERT with Disentangled Attention) uses enhanced attention mechanisms to better capture dependencies in text.

For our current objective, these models are refined end-to-end on labelled instances of both rule-violating and non-violating content. The model learns to recognize linguistic clues, semantic errors, and deviations from community norms that indicate a rule violation throughout training. Because transformer models are extremely successful at picking up minor contextual nuances that traditional natural language processing systems frequently overlook, such as sarcasm, indirect violations, and implicit rule breaches, they are especially useful for this task. This method produces extremely expressive classifiers that can represent the intricacies of conversational input in the style of Reddit discourse and online conversation.

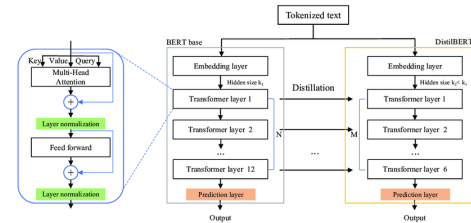


Figure 6: DistilBERT Architecture, adapted from (Silva Barbon and Akabane 2022)

The second method uses LLM embeddings to recast rule-violation detection as a semantic similarity problem. This approach generates dense vector embeddings for comments, rules, positive examples, and negative examples using a big language model rather than fine-tuning a classifier. High-level semantic meaning and relational structure are captured in a shared vector space by these embeddings.

A new post is classified by comparing its embedding to the embedded instances from the training dataset using either vector distance or cosine similarity. If a new post is semantically similar to known rule-violating examples or dissimilar from complying ones, it is likely to violate the rules. Extensions like retrieval-augmented moderation, rule-conditioned Q&A, and explanation generation are made possible by this embedding-based method. This supports zero-shot generalization and is resilient to language and style variations. Additionally, it expands well to big datasets using vector search tools like FAISS or Elasticsearch, avoiding expensive fine-tuning.

Transformer-Based Models

Modern state-of-the-art text categorization systems are built on transformer-based architectures. Transformers use self-attention, which enables the model to capture global context-

tual links between words in a single step, in contrast to recurrent networks, which process text sequentially. For tasks involving long-range dependencies, sensitive semantic clues, and complex linguistic patterns, like toxicity detection, they are therefore very effective and efficient.

Modern Transformer models such as BERT, RoBERTa, DistilBERT, and DeBERTa are trained on billions of tokens before being fine-tuned on a specific downstream task.

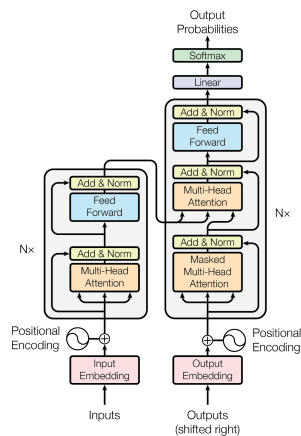


Figure 7: Transformer Architecture

DistilBERT DistilBERT is a distilled, lightweight version of BERT that is 40–60% smaller and faster while maintaining roughly 95% of its language understanding capabilities. As seen in figure 6, it eliminates multiple layers and compresses the representations of the original model using knowledge distillation.

DistilBERT’s capacity to detect delicate linguistic cues like sarcasm, complex insults, implied toxicity, or contextual complexity may be limited by its lower parameter count, despite the fact that it is computationally efficient and simple to train. This explains why, in comparison to the more sophisticated DeBERTa and DistilRoBERTa models in this project, its leaderboard score is much lower (0.4001). DistilBERT is still a solid baseline with good speed-performance trade-offs in spite of this.

DistilRoBERTa RoBERTa, an optimized version of BERT trained with more data, larger batches, and dynamic masking, is distilled into DistilRoBERTa. The model uses distillation to reduce its size and maintain good performance while being more efficient than complete RoBERTa.

DistilRoBERTa outperformed DistilBERT with a public score of 0.55196, nearly matching DeBERTa. This implies that even in condensed form, RoBERTa’s strong token representations and deeper pre-training regime make a significant contribution to the modelling of harmful language.

DeBERTa DeBERTa (Decoding-Enhanced BERT with Disentangled Attention) enhances the standard BERT architecture by dividing word content and word position information into two attention spaces. Additionally, it improves

decoding by applying absolute position signals following self-attention. These developments enable DeBERTa to capture more accurate contextual meaning and deeper semantic linkages. This is useful for subtle or emotionally charged instances of toxicity in the data.

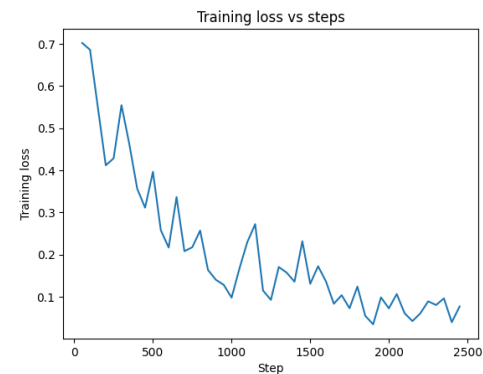


Figure 8: Train Loss vs Steps for DeBERTa

During training, the DeBERTa model displayed rapid convergence, with the evaluation loss falling steadily throughout the first 2000 steps. After this point, the validation loss plateaued, indicating that the model had already extracted most of the significant patterns accessible in the dataset. This plateau is prevalent in big pre-trained Transformer models, which already start with a strong initialization and hence require fewer fine-tuning steps for convergence.

Out of all the Transformer models evaluated in this project, DeBERTa received the highest leaderboard score of 0.56281. This is consistent with its excellent performance on benchmark tasks like GLUE and SuperGLUE, where it usually outperforms variations of BERT and RoBERTa.

LLM Embedding Based Approach

Instead of framing the rule-violation prediction task as a supervised classification issue, this method reframes it as a retrieval-based similarity problem. The approach uses nearest-neighbor reasoning to infer rule violations by embedding Reddit comments into a semantic vector space rather than constructing a model to learn decision boundaries.

Data Preparation The corpus used for similarity search is constructed using the same foundational idea as in earlier preprocessing:

1. All the comments from the training file are included
2. The positive and negative examples from the test set are also added with +1 embedding for the positive (rule-violating) examples and +1 for the negative
3. Every comment in the augmented corpus is embedded using the Qwen3-Embedding-0.6B model, producing high-dimensional dense vector representations to encode contextual meaning
4. A light cleaning step like previous preprocessing is applied to improve embedding consistency by removing noisy formatting, URLs, or irregular tokenization

Inference Pipeline For each validation sample, the following pipeline is established:

1. Compute all the embeddings
2. Retrieve its K nearest neighbours from the corpus belonging to the same rule
3. Compute the score for the embedding with this formula:

$$score = \sum_{i=1}^K (similarity_i \times label_i)$$

4. Convert the score into a probability

Results With a public leaderboard score of **0.87178**, the LLM-based semantic search approach outperformed the transformer baselines by a significant margin. This outcome demonstrates the effectiveness of retrieval-based reasoning in this problem, where a crucial predictive signal is provided by resemblance to known rule-abiding and rule-violating examples. Dense Qwen3 embeddings, rule-specific neighbour filtering, and weighted similarity scoring were found to be an effective yet potent substitute for supervised learning.

The approach not only matched but also significantly outperformed the transformer models without requiring gradient-based training, proving that semantically rich embedding spaces can capture subtle policy-violation patterns found in community moderation data when combined with carefully thought-out retrieval strategies.

Evaluation

There are a number of reasons why the semantic-search embedding method outperformed the transformer models. Firstly, determining whether a statement breaks a particular community rule is a highly contextual problem that relies more on semantic resemblance to existing cases than on general linguistic patterns. Conventional fine-tuning may not adequately reflect the vast diversity of rule-contexts among subreddits since it relies on the model to learn these subtleties from a small amount of task-specific data.

On the other hand, rich semantic relations obtained from extensive pre-training are already encoded by dense LLM embeddings (like Qwen3-embedding-0.6b). Nearest-neighbour retrieval is an intrinsically robust baseline for this kind of classification because when comments are compared in this embedding space, violations and non-violations naturally cluster according to meaning.

Table 1: Public and Private Scores for the Models Used

Model	Public Score	Private Score
DistilBERT	0.40012	0.41106
DistilRoBERTa	0.55196	0.53920
DeBERTa 2	0.56281	0.56276
Qwen3-Embedding-0.6B	0.87178	0.87873

Because the competition uses column-averaged AUC, our model outputs probability estimates rather than discrete labels. These probabilities represent the model’s confidence

that a particular rule was violated and allow the evaluator to compute the ROC curves and AUC values for each rule.

Finally, the LLM’s robustness and simplicity are major advantages. Due to the lack of a training loop, it avoids overfitting, catastrophic forgetting, and optimization instability, which can be prevalent problems that impact the transformer models (such as plateauing validation loss in DeBERTa). Instead, the embedding-based pipeline consistently achieves high generalization performance by utilizing deterministic similarity scoring and reliable, pre-trained language representations.

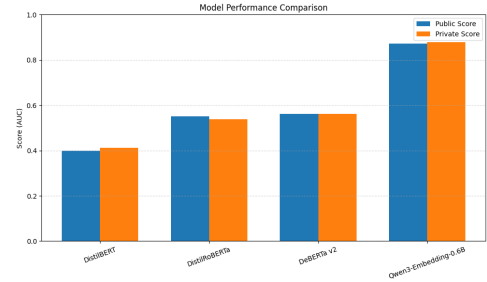


Figure 9: Model-wise Score Distribution

Future Scope

With its lightweight architecture reaching 0.87+ AUC, the Qwen3-Embedding-0.6B model produces surprisingly good results. However, there is much room for improvement by employing larger and higher-capacity LLMs. These models exhibit stronger semantic comprehension, improved contextual reasoning, and richer representations, which can directly improve performance in tasks like rule-violation detection that require complex text classification objectives.

Using Large Language Models (LLMs) like GPT-4, LLaMA-3, or Mistral in the moderation process is a strong future approach. LLMs can be employed in a judgment framework, where they assess whether a post breaks a rule and offer an explanation, rather than depending exclusively on classification models. The speed of transformer classifiers and the sophisticated reasoning power of LLMs could be combined in this hybrid configuration. This approach is consistent with recent research on "LLM-as-a-Judge" systems that are employed in pipelines for content moderation and safety.

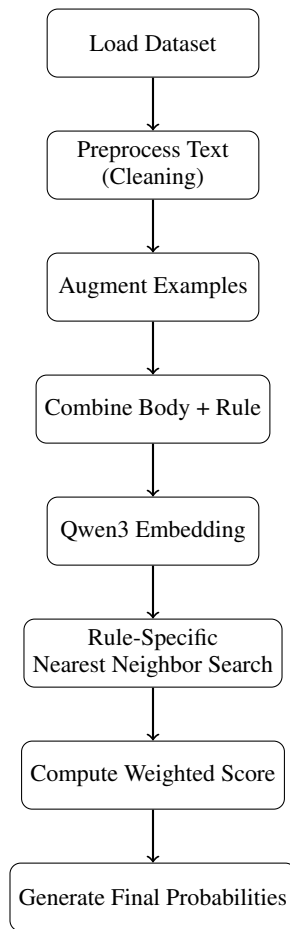


Figure 10: Flowchart of the Semantic Search-based Classification Pipeline.

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